

Client Governance CRM + Project Tracking Platform

Complete Software Architecture & Development Documentation

Prepared in Simple English

1) Project Overview

Product Vision

This product is a **multi-role web application** that helps the company manage complete client lifecycle operations.

It combines: - CRM - project progress tracking - client portal - founder dashboard - team productivity - cost and profitability tracking - weekly and monthly governance

The current Excel workbook will become the **main product modules/pages**.

2) Main Goals of the Product

The platform should solve these business problems:

For Founder

- overall company health
- total active clients
- completed projects
- revenue and gross margin
- delayed projects
- client review visibility
- team productivity
- company-wide dashboard

For Team / Interns

- add new companies after deal closure
- update milestones
- update AI use cases

- update weekly reports
- add executive review notes
- maintain training trackers
- see own monthly work contribution

For Client

- secure login
 - see only their own company data
 - project completion percentage
 - milestone roadmap
 - completed deliverables
 - in-progress work
 - weekly updates
 - review notes
-

3) User Roles & Permissions

1. Founder (Super Admin)

Full access to all companies and all modules.

Permissions

- manage all users
- create companies
- assign team members
- view all dashboards
- cost and revenue access
- see client health
- see team productivity
- create client accounts
- access reports

2. Team / Intern

Operational role.

Permissions

- add company data
- update project progress
- update training sheets
- add weekly email reports
- add monthly review notes

- manage assigned clients
- view own productivity

3. Client

Restricted access.

Permissions

- only own company
 - project progress
 - milestone status
 - use case completion
 - training completion
 - weekly updates
 - meeting notes
-

4) Final Tech Stack

Frontend

- **React.js**
- **Vite**
- **Tailwind CSS**
- **ShadCN UI**
- **Recharts** for graphs
- **React Router**
- **Axios**

Backend

- **Node.js**
- **Express.js**
- REST APIs

Database

- **PostgreSQL**

Authentication

- JWT based authentication
- role-based access control
- bcrypt password hashing

Deployment

- Frontend → Vercel
- Backend → Render / Railway
- Database → Neon PostgreSQL / Supabase PostgreSQL

Optional Later Additions

- Redis caching
 - AWS S3 for documents
 - email automation
 - audit logs
-

5) High-Level Product Architecture

```
Frontend (React)
  ↓
API Layer (Express)
  ↓
Business Logic Layer
  ↓
PostgreSQL Database
  ↓
Reports / Analytics / Notifications
```

Architecture Flow

Step 1: User Login

User logs in using email and password.

Step 2: JWT Token

Backend verifies credentials and sends token.

Step 3: Role-Based Dashboard

System redirects based on role: - founder dashboard - team dashboard - client dashboard

Step 4: CRUD Operations

All page updates go through APIs.

Step 5: Database Update

Postgres stores all data safely.

6) Final Pages / Modules (7–8 Main Pages)

These pages come directly from the Excel workbook + founder requirements.

1. Dashboard Page

Different for each role.

Founder Dashboard Widgets

- active companies
- completed projects
- delayed projects
- total revenue
- gross margin
- monthly burn
- team productivity

Client Dashboard Widgets

- project completion %
 - current milestone
 - completed modules
 - next review date
-

2. Company Management Page

This is the CRM company master.

Features

- add company
- edit company
- assign team member
- client details
- contract details
- engagement start date
- status

3. AI Use Cases Tracker Page

This is the main delivery tracking page.

Fields

- use case name
 - department
 - complexity
 - current phase
 - UAT status
 - go-live date
 - comments
 - progress percentage
-

4. AI Training Tracker Page

Tracks all client training activities.

Fields

- training title
 - duration
 - trainer
 - audience
 - date
 - completion status
 - feedback
-

5. Weekly Execution / Weekly Email Report Page

This replaces the weekly email report Excel sheet.

Features

- weekly summary
 - completed tasks
 - next week plan
 - blockers
 - email-ready report
 - client visible notes
-

6. Monthly Executive Review Page

For founder + client leadership governance.

Features

- meeting agenda
 - attendees
 - MOM notes
 - action items
 - next review date
 - escalation risks
-

7. Cost Tracker Page

Finance visibility page.

Fields

- deal value
 - revenue accrued
 - FTE cost
 - freelancer cost
 - software cost
 - travel cost
 - total cost
 - gross margin
 - GM %
-

8. Team Productivity / Monthly Work Page

Tracks team performance.

Features

- assigned clients
 - completed updates
 - milestones completed
 - weekly reports submitted
 - use cases deployed
 - monthly score
-

7) PostgreSQL Database Design

This is the most important backend layer.

1. users

Stores login credentials.

```
id
name
email
password_hash
role
created_at
updated_at
```

2. companies

Stores company/client details.

```
id
company_name
industry
country
status
contract_value
start_date
created_by
created_at
```

3. company_team_members

Mapping between company and internal team.

```
id
company_id
```

```
user_id
assigned_at
```

4. projects

Main project tracking.

```
id
company_id
project_name
completion_percentage
current_phase
status
start_date
expected_end_date
```

5. use_cases

Tracks AI use cases.

```
id
project_id
use_case_name
department
complexity
status
uat_date
go_live_date
comments
```

6. trainings

```
id
company_id
training_name
trainer_name
duration
status
```

```
training_date  
feedback
```

7. weekly_reports

```
id  
company_id  
week_start  
summary  
completed_work  
next_plan  
blockers  
created_by
```

8. monthly_reviews

```
id  
company_id  
review_date  
agenda  
mom_notes  
action_items  
next_review_date
```

9. costs

```
id  
company_id  
revenue  
fte_cost  
freelancer_cost  
license_cost  
travel_cost  
total_cost  
gross_margin  
margin_percentage  
month
```

10. productivity_logs

```
id
user_id
company_id
tasks_completed
reports_submitted
milestones_closed
month
score
```

8) API Architecture (Backend Routes)

Auth APIs

- POST /auth/login
- POST /auth/register
- GET /auth/me

Company APIs

- POST /companies
- GET /companies
- GET /companies/:id
- PUT /companies/:id

Project APIs

- POST /projects
- PUT /projects/:id/progress
- GET /projects/company/:companyId

Use Case APIs

- POST /usecases
- PUT /usecases/:id
- GET /usecases/project/:projectId

Weekly Reports APIs

- POST /weekly-reports

- GET /weekly-reports/company/:companyId

Review APIs

- POST /monthly-reviews
- GET /monthly-reviews/company/:companyId

Cost APIs

- POST /costs
 - GET /costs/company/:companyId
-

9) Development Folder Structure

```
client/  
├── src/  
│   ├── pages/  
│   ├── components/  
│   ├── services/  
│   ├── hooks/  
│   └── layouts/  
  
server/  
├── controllers/  
├── routes/  
├── middleware/  
├── models/  
├── config/  
└── utils/
```

10) Development Execution Plan

Phase 1

- auth
- role-based login
- dashboards

Phase 2

- company CRM module

- project module
- use case tracker

Phase 3

- training
- weekly reports
- executive reviews

Phase 4

- cost tracker
- productivity dashboard
- analytics charts

Phase 5

- deployment
- testing
- founder demo

11) Recommended UI Flow

Founder Flow

Login → Dashboard → Company → Project → Cost → Team Analytics

Team Flow

Login → Assigned Clients → Update Progress → Weekly Reports

Client Flow

Login → Own Dashboard → Progress → Weekly Updates → Review Notes

12) Final Founder-Ready Product Summary

This product is a:

Multi-role CRM + Client Governance + Project Tracking + Client Portal Platform

It replaces: - scattered Excel files - manual reporting - no founder visibility - no client transparency - no team productivity tracking

with a single centralized web application.

This is a **real SaaS-grade internal company platform**.

13) Suggested Future Enhancements

- automated weekly email sender
 - AI-generated MOM summaries
 - risk prediction dashboard
 - SLA alerts
 - WhatsApp reminders
 - client satisfaction score
 - file uploads
 - invoice tracking
 - resource utilization heatmap
-

14) Recommended Start

The best build order:

1. Auth + role routing
2. Dashboard
3. Company CRUD
4. Project progress
5. Weekly reports
6. Client portal
7. Cost tracker

This will help us build fast and demo early.